Noise Sensitivity in Radar Ship Tracking

A Comparison of Linear and Nonlinear Kalman Filters

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Estimation and Filtering | Project 2

Abstract. We compare a constant-velocity Kalman filter with converted radar measurements against extended, unscented, cubature, and square-root cubature Kalman filters for two-dimensional ship tracking. A paired Monte Carlo experiment varies bearing and range noise separately, keeping the process covariance fixed and the other measurement-noise parameter nominal. Each condition uses 30 trajectories and 800 radar scans, with the true noise parameters supplied to every filter. At nominal noise, position errors are similar, while velocity errors differ substantially. At large bearing and range noise the nonlinear trackers outperform the baseline, but their relative ranking depends on the measurement-noise component and on whether position or velocity error is considered. The study reports reproducible sensitivity results and documents the confounding effect of motion-model mismatch.

1 Introduction and related work

Radar tracking requires inferring position and velocity from noisy observations. The Kalman filter provides a recursive linear minimum-mean-square-error estimator when its model and second-order statistics are correctly specified; in the linear Gaussian setting it also recovers the exact conditional mean [1]. Radar range and bearing are nonlinear functions of Cartesian position, so a linear state model alone does not make the observation problem linear.

Two common responses are to linearize the observation map or approximate its transformed moments. Julier and Uhlmann explain the unscented approach and its relationship to local linearization [2]. Arasaratnam and Haykin introduce a spherical-radial cubature rule and a square-root implementation [3]. A different route converts radar observations into Cartesian coordinates before a linear update. The work of Lerro and Bar-Shalom [4] and the later study by Spitzmiller and Adhami [5] motivate careful treatment of conversion bias and covariance. Pei gives a recent comparison of converted-measurement methods [6].

This course project adapts the maneuvering-ship scenario in Section 6.1 of Arasaratnam's thesis [7]. It adds a constant-velocity converted-measurement baseline and two measurement-noise sensitivity sweeps with fixed process covariance. The contribution is an experimental comparison, not a new filtering algorithm or an exact reproduction of the thesis. In particular, the baseline does not use the turning model available to the nonlinear filters; this structural difference is part of the comparison.

2 State and observation models

2.1 Discrete ship dynamics

Let j index integration substeps and let k index radar scans. The state is $\boldsymbol{x}_j = \left(\boldsymbol{p}_j^T, \boldsymbol{v}_j^T\right)^T$, with two-dimensional position $\boldsymbol{p}_j$ and velocity $\boldsymbol{v}_j$. The implementation advances

$$\begin{aligned} \boldsymbol{p}_{j+1} &= \boldsymbol{p}_j + \delta \boldsymbol{v}_j + q_p \boldsymbol{\varepsilon}_{p,j}, \\ \boldsymbol{v}_{j+1} &= \boldsymbol{v}_j + \boldsymbol{c}\big(\boldsymbol{p}_j, \boldsymbol{v}_j\big) + q_v \boldsymbol{\varepsilon}_{v,j}. \end{aligned} \tag{1}$$

The independent noise vectors have standard normal distributions. The deterministic turning increment is

$$\boldsymbol{c}(\boldsymbol{p}, \boldsymbol{v}) = \begin{cases} -\lambda \dfrac{\boldsymbol{p}}{\|\boldsymbol{p}\|} & \\ \qquad \|\boldsymbol{p}\| \geq r_s \quad \text{and} \quad \boldsymbol{p}^T \boldsymbol{v} \geq 0 & \\ \mathbf{0} & \\ \qquad \text{otherwise.} & \end{cases} \tag{2}$$

Here $r_s = 5$ and $\lambda = 2$. The turning term is a **discrete velocity increment**, exactly as coded; it is not multiplied by δ again. The additive substep covariance is

$$Q = \operatorname{diag}\big(q_p^2, q_p^2, q_v^2, q_v^2\big). \tag{3}$$

We use $\delta = 0.01$ and five substeps per scan, giving a radar interval $\Delta = 0.05$. The thesis uses normalized space and time units. To retain the existing plot labels, this project assigns one space unit to 1 m and one time unit to 1 s. These are display conventions, not an empirically calibrated vessel model; comparisons are unchanged by retaining normalized units.

2.2 Radar observations

A stationary radar at the origin observes $\boldsymbol{x}_{5k}$:

$$\boldsymbol{z}_k = h(\boldsymbol{x}_{5k}) + \boldsymbol{\nu}_k, \quad h(\boldsymbol{x}) = \begin{pmatrix} \sqrt{p_x^2 + p_y^2} \\ \operatorname{atan2}(p_y, p_x) \end{pmatrix}, \tag{4}$$

$$\boldsymbol{\nu}_k \sim \mathcal{N}(\mathbf{0}, R), \quad R = \operatorname{diag}(\sigma_r^2, \sigma_\theta^2). \tag{5}$$

Measurement noise, process noise, and initial states are independent. Bearing standard deviations are specified in degrees and converted to radians before constructing R. Bearing residuals are wrapped to $[-\pi, \pi)$. The additive Gaussian range model is retained without clipping, so near the origin a noisy range can be negative.

3 Estimation methods

3.1 Linear constant-velocity baseline

The KF-CV baseline replaces the turning increment by zero. For the state order $\left[p_x, p_y, v_x, v_y\right]^T$, its substep transition and Cartesian observation matrices are

$$F = \begin{pmatrix} 1 & 0 & \delta & 0 \\ 0 & 1 & 0 & \delta \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad H_c = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \tag{6}$$

Five substeps are aggregated exactly: $A = F^5$ and $Q_\Delta = \sum_{\ell=0}^{4} F^\ell Q \left(F^\ell\right)^T$. The prediction is $\hat{\boldsymbol{x}}_k^- = A\hat{\boldsymbol{x}}_{k-1}^+$ and $P_k^- = AP_{k-1}^+A^T + Q_\Delta$.

Raw Cartesian conversion is biased because $\mathbb{E}[\cos(\nu_\theta)] = \exp\left(-\frac{\sigma_\theta^2}{2}\right)$. With $a = \sigma_\theta^2$, we use the multiplicatively unbiased observation

$$\boldsymbol{z}_{c,k} = \exp\left(\frac{a}{2}\right) r_{m,k} \begin{pmatrix} \cos(\theta_{m,k}) \\ \sin(\theta_{m,k}) \end{pmatrix}. \tag{7}$$

At a fixed true range ρ, its radial and tangential error variances are

$$\begin{aligned} d_r &= (\rho^2 + \sigma_r^2)\cosh(a) - \rho^2, \\ d_t &= (\rho^2 + \sigma_r^2)\sinh(a). \end{aligned} \tag{8}$$

These expressions follow from the Gaussian sine and cosine moments. Rotating $\operatorname{diag}(d_r, d_t)$ into Cartesian axes gives the covariance at a fixed true position. Since truth is unavailable, the implementation substitutes its predicted position. This is a plug-in approximation, not the full statistically consistent converted-measurement algorithm of [4] or [5]. Although F and H_c are fixed, the converted covariance is prediction-dependent, and converted errors are non-Gaussian. Thus KF-CV is a linear-model baseline with an approximate radar interface, not an exact Gaussian solution to Equation 4.

3.2 Common correction and extended Kalman filter

The nonlinear methods maintain a predicted state, predicted observation, innovation covariance S_k, and state-observation cross-covariance C_k. Their correction uses

$$\begin{aligned} K_k &= C_k S_k^{-1}, \\ \hat{\boldsymbol{x}}_k^+ &= \hat{\boldsymbol{x}}_k^- + K_k(\boldsymbol{z}_k - \hat{\boldsymbol{z}}_k), \\ P_k^+ &= P_k^- - K_k S_k K_k^T. \end{aligned} \tag{9}$$

For KF-CV the observation and covariance are Cartesian. For the nonlinear filters, the bearing component of the innovation is wrapped. The implementation uses the Joseph covariance form for KF-CV and EKF.

The EKF propagates $\hat{\boldsymbol{x}}_{j+1} = f\left(\hat{\boldsymbol{x}}_j\right)$ and $P_{j+1} = F_j P_j F_j^T + Q$, evaluating the dynamics Jacobian at the state **before** propagation. For nonzero range, the observation Jacobian is

$$H_k = \begin{pmatrix} \frac{p_x}{\rho} & \frac{p_y}{\rho} & 0 & 0 \\ -\frac{p_y}{\rho^2} & \frac{p_x}{\rho^2} & 0 & 0 \end{pmatrix}. \tag{10}$$

On a fixed active branch, $F_j = \begin{pmatrix} I_2 & \delta I_2 \\ J_c & I_2 \end{pmatrix}$ with $J_c = -\lambda\left(\frac{I_2}{\rho} - \boldsymbol{p}\frac{\boldsymbol{p}^T}{\rho^3}\right)$; on the inactive branch $J_c = 0$. The switching surface is not differentiable, so local Gaussian approximations are especially restrictive near turns.

3.3 Unscented, cubature, and square-root filters

The implemented UKF uses $n = 4$, $\alpha = 1$, $\beta = 2$, and $\kappa = 1$, so $\lambda_U = \alpha^2(n + \kappa) - n = 1$. It places $2n + 1 = 9$ sigma points at the mean and at symmetric offsets given by columns of $\sqrt{(n + \lambda_U)P}$. The mean weights are $W_0^m = 0.2$ and $W_i^m = 0.1$ for outer points; the central covariance weight is $W_0^c = 2.2$. Points are transformed through the actual nonlinear maps, then recombined [2].

For the CKF, if $P = LL^T$, the eight points and weights are

$$\boldsymbol{\chi}_{i,\pm} = \hat{\boldsymbol{x}} \pm \sqrt{n} L \boldsymbol{e}_i, \quad w_{i,\pm} = \frac{1}{2n}, \quad i = 1, \ldots, n. \tag{11}$$

For a transformation g, their weighted mean and centered outer products approximate its mean and covariance; additive process or measurement covariance is then included. The CKF and its square-root variant use this same cubature rule [3]. SCKF carries covariance factors and uses QR factorizations and triangular solves rather than explicit covariance subtraction. It changes numerical representation, not the underlying integration rule.

Both UKF and CKF regenerate points after adding process noise. Their predicted bearing uses a weighted circular mean, $\operatorname{atan2}\left(\sum_i w_i \sin(\theta_i), \sum_i w_i \cos(\theta_i)\right)$. Equal positive CKF weights do not establish exact nonlinear moments, strict positive definiteness in every numerical situation, or universal accuracy superiority.

4 Experimental design

Two sweeps vary σ_θ and σ_r separately. The bearing sweep holds range SD at 0.01 m, and the range sweep holds bearing SD at 0.5 degrees. Throughout both sweeps, $q_p = q_v = 0.01$ and $Q = 10^{-4} I_4$ remain fixed. Every filter receives the true process and polar-noise parameters; KF-CV still approximates the converted Cartesian covariance as described in Section 3.1. This is a study of **measurement-noise intensity** with known parameters, not robustness to incorrectly assumed Q or R.

Parameter	Nominal	Sweep values / fixed setting
σ_θ (degrees)	0.5	0.5, 2, 5, 10, 20, 30
σ_r (m)	0.01	0.01, 0.02, 0.05, 0.1, 0.2, 0.5
q_p (m)	0.01	Fixed in both sweeps
q_v (m/s)	0.01	Fixed in both sweeps

Table 1: One standard deviation varies at a time. Process parameters are SDs of discrete substep increments, not diffusion intensities. All other entries retain their nominal values. Units follow Section 2.1.

Each condition uses 30 trajectories and 800 scans with seed 20261026. True initial states are drawn uniformly from $[-0.5, 0.5]^4$; every filter starts from $\hat{\boldsymbol{x}}_0 = (1, 1, 1, 1)^T$ and $P_0 = 10 I_4$. The same initial states and standardized Gaussian process and measurement draws are reused across conditions. All conditions therefore preserve the same true trajectories. Only the selected radar-noise amplitude changes. Within each condition, all methods receive the same observations. The nominal condition is shared across sweeps, yielding 11 unique configurations and 12 displayed points.

For trajectory i, let $\boldsymbol{e}_{p,k}^i$ denote its two-dimensional position error at radar scan k. We compute

$$\mathrm{RMSE}_{p(k)} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left\| \boldsymbol{e}_{p,k}^i \right\|^2}, \quad \overline{\mathrm{RMSE}}_p = \frac{1}{K - B} \sum_{k=B+1}^{K} \mathrm{RMSE}_{p(k)}. \tag{12}$$

Velocity uses the same construction. Here $N = 30$, $K = 800$, and $B = 300$. Thus summary results average per-scan RMSE over scans 301-800, corresponding to 15.05-40 s under the display convention. They are not obtained by pooling time and trajectory errors before taking a square root. A failed or non-finite track invalidates the whole method-condition pair; no survivor-only averages are used. All methods completed every track in every condition.

Elapsed filtering time includes the baseline's conversion and excludes simulation and plotting. It is averaged over the tracks in one batch, not over repeated timing experiments. The execution environment was Darwin arm64, Python 3.14.4, NumPy 2.5.1. Eleven numerical tests cover conversion moments, process covariance propagation, parameter isolation, paired random draws, and failure handling. The current dataset and simulation source hashes are stored together in `noise-sensitivity.json`.

5 Results

5.1 Nominal noise

Method	Position (m)	Velocity (m/s)	Time (ms/track)
KF-CV	0.0225	0.4378	19
EKF	0.0222	0.1818	66.2
UKF	0.0223	0.1684	147.8
CKF	0.0223	0.162	144.1
SCKF	0.0223	0.162	183.2

Table 2: Nominal-noise results: mean per-scan RMSE after initialization and mean filtering time per 800-scan trajectory. Values are read directly from the current experiment JSON.

At nominal noise, position RMSE is nearly identical across methods (Table 2). Velocity is more discriminating: KF-CV has error 0.4378 m/s, compared with 0.1818 m/s for EKF, 0.1684 m/s for UKF, and 0.162 m/s for CKF. KF-CV is fastest in this implementation, while SCKF costs more than CKF. Aggregated linear prediction, the number of point transformations, and implementation details all affect these timings.

5.2 Measurement-noise sensitivity

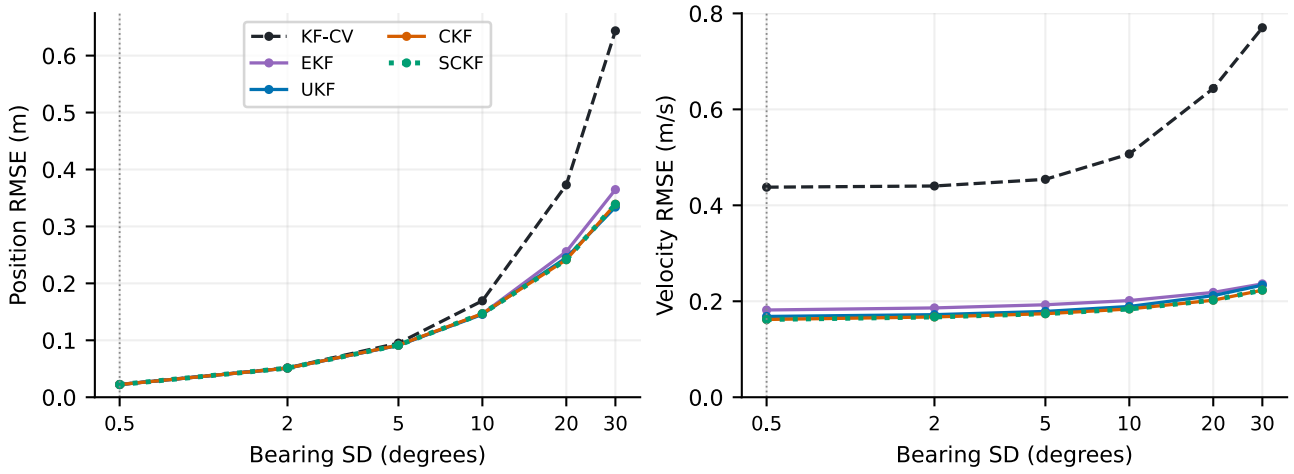


Figure 1: Bearing-noise sweep. Range and both process noises remain nominal. Curves show mean per-scan RMSE after initialization; the dotted vertical line marks nominal noise. SCKF overlaps CKF at the plotted precision.

At 30-degree bearing noise, KF-CV position RMSE is 0.6435 m, versus 0.339 m for CKF (Figure 1). This is a 47.3% reduction in this ensemble. UKF has slightly lower position error than CKF at that endpoint, whereas CKF has lower velocity error. The experiment therefore does not establish a single winner even among the nonlinear methods.

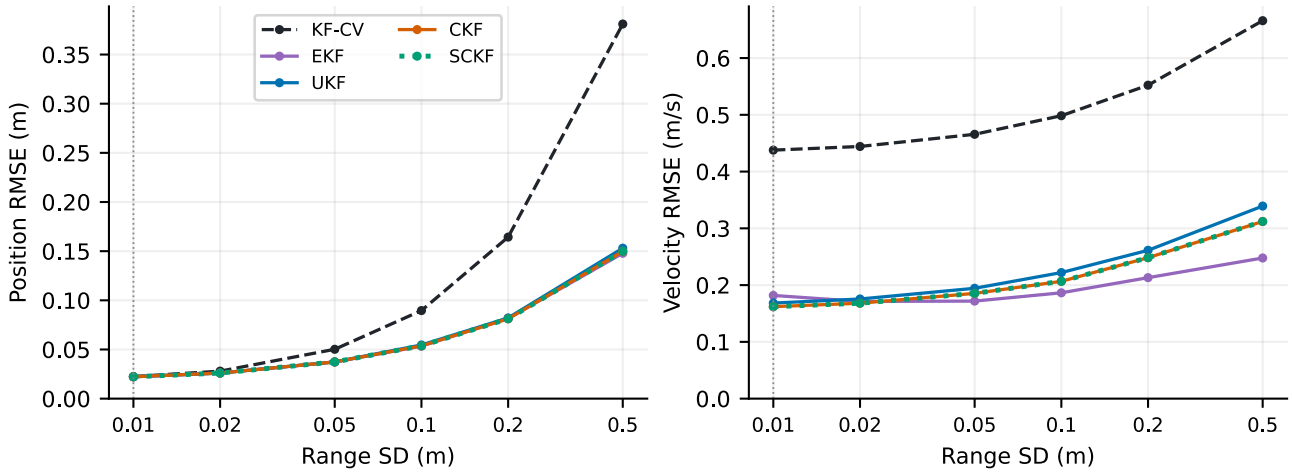


Figure 2: Range-noise sweep with bearing standard deviation fixed at 0.5 degrees and both process noises nominal. The Gaussian range model is not truncated near zero.

At range SD 0.5 m, KF-CV position RMSE is 0.381 m and CKF reaches 0.1498 m. However, EKF has the smallest endpoint velocity RMSE among these nonlinear methods: 0.2478 m/s, versus 0.3121 m/s for CKF (Figure 2). Increasing range noise does not simply reproduce the bearing-noise ranking.

6 Discussion and limitations

The baseline has linear motion while the other filters know the nonlinear turn. Observed differences therefore combine motion-model mismatch, Cartesian conversion statistics, and nonlinear moment approximations. A further experiment using constant-velocity truth for every method would be needed to isolate measurement nonlinearity. Likewise, no parameter search was performed to build an optimally tuned linear maneuvering tracker.

The switch in Equation 2 is nonsmooth, the initial state prior is deliberately imperfect, and the ensemble contains only 30 paired trajectories. No confidence intervals, hypothesis tests, normalized estimation error squared (NEES), or innovation-consistency analysis have been computed. Small numerical differences should not be described as statistically significant. The results are synthetic and are not validated against measured vessel tracks.

SCKF and CKF agree at displayed precision, as expected from their common cubature rule. This double-precision experiment does not establish a practical stability advantage for SCKF; that would require a controlled numerical stress or reduced-precision study. Finally, all noises remain white Gaussian, with known covariances. Coloured noise and covariance misspecification are separate extensions rather than outcomes demonstrated here.

7 Conclusion

The paired experiment provides the linear KF baseline and comparison curves required by the project while varying bearing and range noise independently at a fixed process covariance. Nonlinear filters reduce the baseline's velocity error in the tested conditions, but position rankings depend on the noise component and level. CKF and SCKF give the same displayed accuracy, while SCKF incurs higher measured runtime. The next revision should strengthen model-controlled comparisons and uncertainty assessment before making broader claims about filter superiority.

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